Basic Principles of Computed Tomography Physics

Detection of X-Rays

1-Detection of x-rays is accomplished by the use of special materials that convert the high energies (tens of keV) of the x-ray quantum into lower energy forms, such as optical photons or electron-hole pairs, which have energies of a few electron volts.

2- In this down-conversion, many secondary quanta are generated, typically thousands per primary quanta.

3- The detector materials, such as phosphors, scintillating ceramics, or pressurized xenon gas, ultimately produce an electrical current or voltage.

4-Electronic amplifiers condition this signal, and an analog-to-digital converter converts it into a digital number.

5-The range of signals produced in tomography is large, varying from a scan of air (no attenuation, or 100% transmission) to that of a large patient with metal implants

Why X- and gamma radiations are useful for medical imaging?

* The main reason is their penetrating ability.

- When they are directed into an object, some of the photons are absorbed or scattered, whereas others completely penetrate the object.

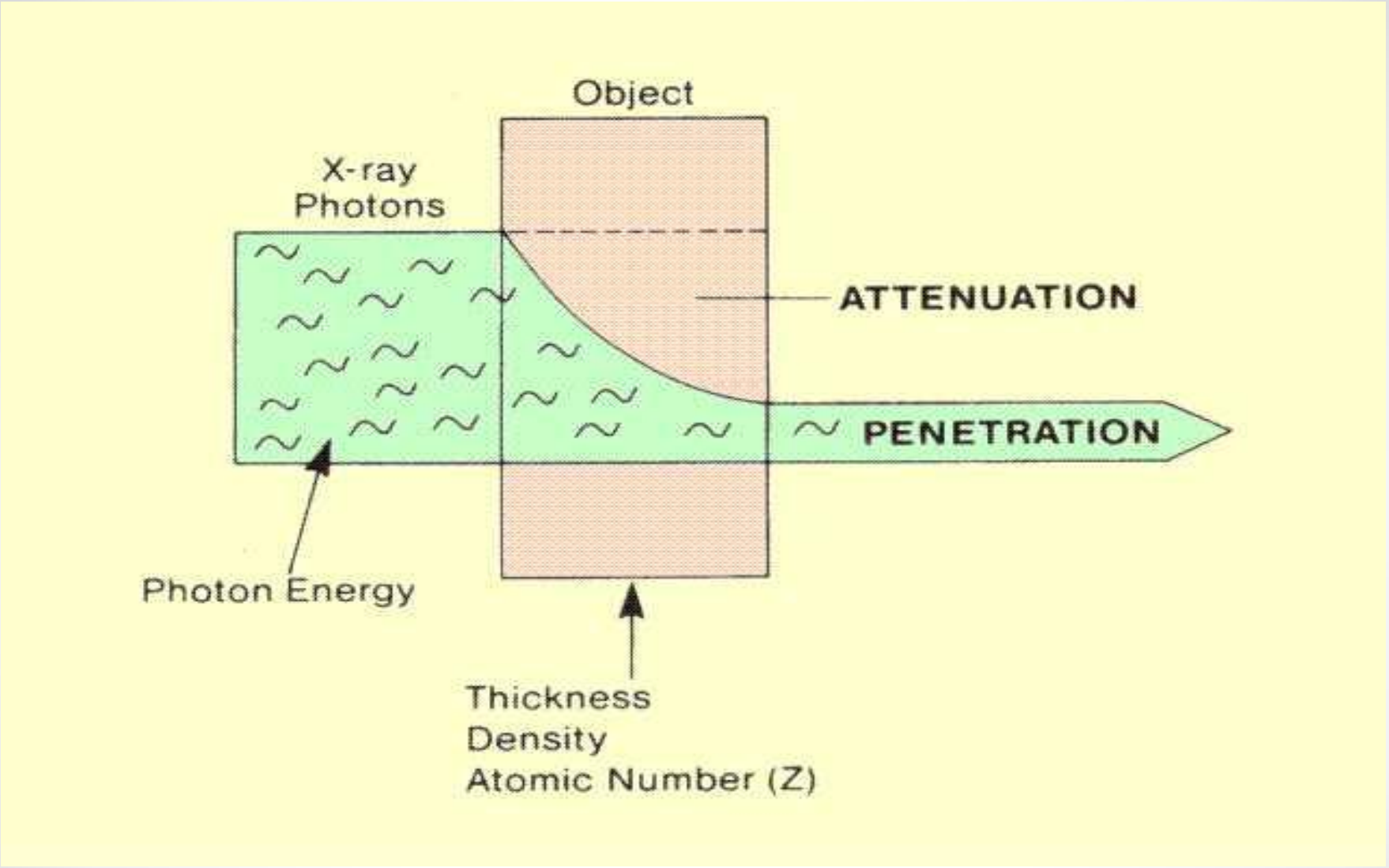
- The penetration can be expressed as the fraction of radiation passing through the object.

- Penetration is the inverse of attenuation.

- The amount of penetration depends on:

1- The energy of the individual photons

2-The atomic number, density, and thickness of the object, as illustrated below.



Factors that affect the penetration of radiation through a specific object

Important notes:

1- The probability of photons interacting, especially with the photoelectric effect, is related to their energy.

2- Increasing photon energy generally decreases the probability of interactions (attenuation) and, therefore, increases penetration.

3- As a rule, high-energy photons are more penetrating than low-energy photons.

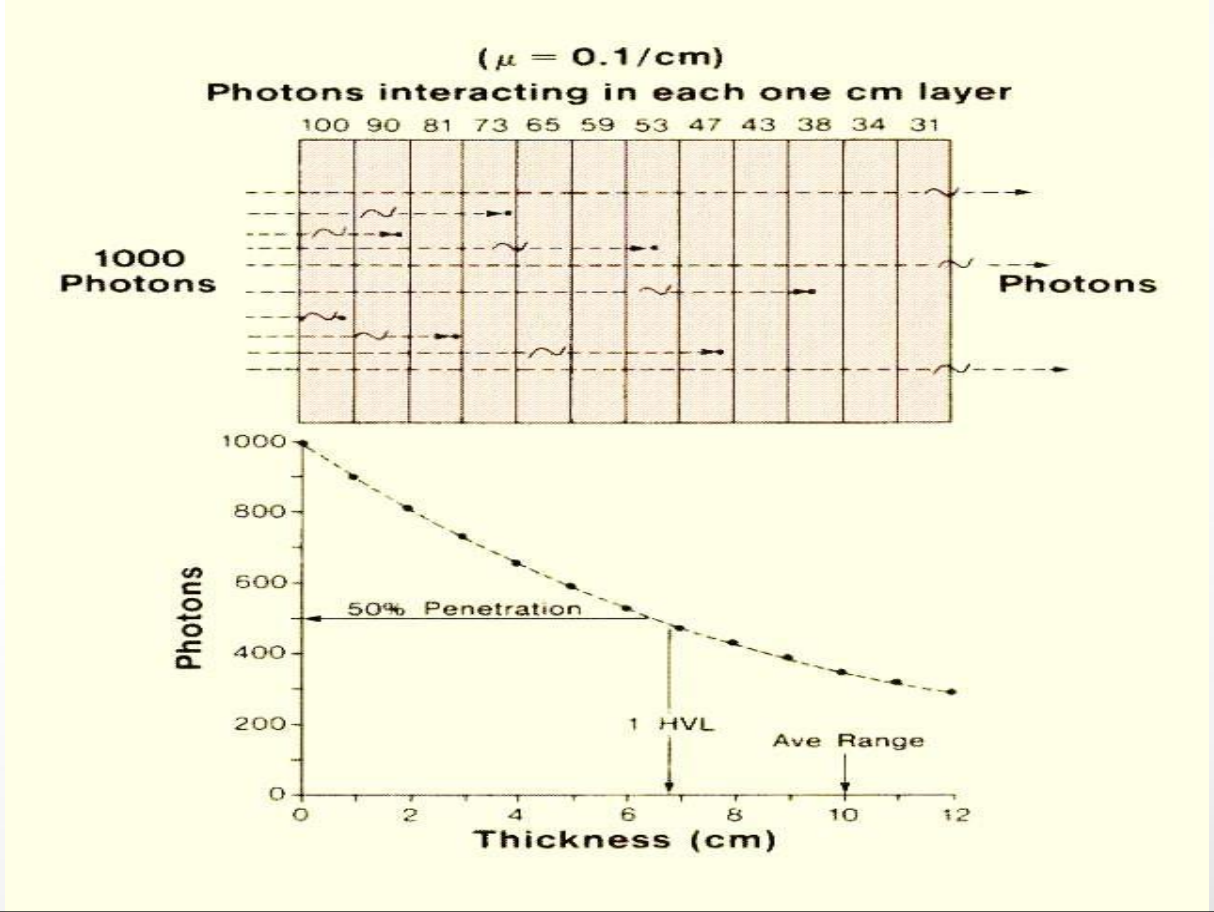
PHOTON RANGE

Photon range is the range, or distance, traveled by the individual photons before they are absorbed or scattered. When photons enter an object, they travel some distance before interacting. This distance can be considered the range of the individual photons.

Note that

A characteristic of radiation is that all photons do not have the same range, even when they have the same energy. In fact, there is no way to predict the range of a specific photon.

Let us consider a group of mono-energetic photons entering an object, as shown below. Some of the photons travel a relatively short distance before interacting, whereas others pass through or penetrate the object. If we count the number of photons penetrating through each thickness of material, we begin to see a fundamental characteristic of photon penetration. The relationship between the number of photons reaching a specific point and the thickness of the material to that point is exponential.



* In a given situation a group of photons have different individual ranges which, when considered together, produce an average range for the group.

The average range is the average distance traveled by the photons before they interact.

- Very few photons travel a distance exactly equal to the average range.

- The average range of a group of photons is inversely related to the attenuation rate.

-Increasing the rate of attenuation by changing photon energy or the type of material decreases the average range of photons.

- Actually, the average photon range is equal to the reciprocal of the attenuation coefficient (μ):

Average Range (cm) =1/Attenuation Coefficient (cm^{-1})